Chronicle

Egocentric Episodic Memory Graph

Condensed technical specification — learn it in one sitting

Derived from the Chronicle Technical Specification and the shipped codebase

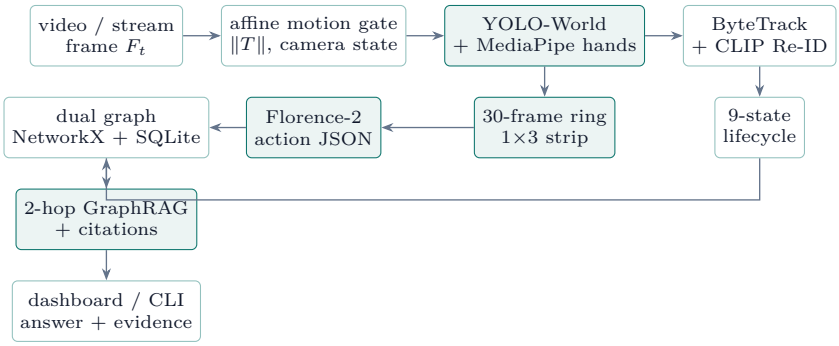
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TL;DR. Chronicle turns first-person video into a *queryable episodic memory*, not a transcript. It samples a video, gates heavy models behind motion and geometric events, writes a dual (background + foreground) spatio-temporal graph to SQLite, and answers natural-language questions from a 2-hop subgraph with timestamped citations. It runs offline with deterministic fallbacks; a local Qwen model is an optional prose layer. A 45-second real clip's memory graph fits in roughly **0.4 MB**, and deterministic graph queries return in **1–5 ms**.

1 The idea in one page

Most video assistants retrieve *clips* or captions. Chronicle instead maintains *object identity over time*: where an object was, when it moved, what it rested on, and who interacted with it. The graph is the source of truth; the language model only phrases the answer.

The pipeline is event-driven. A frame is not sent to a vision-language model unless geometry, motion, or a state transition says something interesting happened. That single design choice is what makes the system fast enough to feel live and small enough to keep a whole episode on disk.



Four modules, four jobs.

- **Perception**: motion gating, open-vocabulary detection, hand landmarks, tracking and Re-ID.
- **Memory**: a 9-state lifecycle, support/containment reasoning, and a durable dual graph.
- **Semantics**: temporal strips, Florence-2 action verification, confidence calibration, GraphRAG.
- **Interface**: CLI, two dashboards, live ingest, uploads, snapshot/CSV export, query console.

2 Data model: two graphs, one episode

Graph	What it stores
Background	Invariant anchors: zones (Kitchen, Work Desk) and fixed furniture (countertop, table, sofa). Keyframes are matched by CLIP place recognition at cosine ≥ 0.85.
Foreground	Episodic objects (keys, phone, mug) with persistent IDs, states, bounding boxes and time-intervaled relations.

Edges. Spatial (`RESTING_ON`, `INSIDE`, `IN_ZONE`), temporal (state transitions with start/end timestamps), and action multi-edges (`USER -[PICKED_UP @ t]-> KEYS`). The *terminal relational anchor* is never deleted: a long-lost object still answers "where was it last?".

2.1 The 9-state lifecycle

State	Deterministic trigger (wall-clock, not frame count)
`NEW`	Detection with no track match and Re-ID cosine < 0.82.
`VISIBLE_STATIC`	Relative velocity ≈ 0 for > 0.5 s and support edge verified.
`INTERACTED`	Hand landmark containment or IoS(hand, object) > 0.60 for ≥ 0.3 s.
`VISIBLE_MOVING`	Object moves independently, hand IoS < 0.10.
`HELD`	Track persists, IoS > 0.50, object centroid follows hand velocity.
`CONTAINED`	Object collapses inside a container hull, IoS > 0.85.
`OCCLUDED`	Track lost while the camera is still and the last box overlaps an occluder.
`OUT_OF_VIEW`	Track lost during a camera pan; spatial anchor retained.
`LOST`	Out of view for > 600 s; node goes dormant, terminal anchor preserved.

Why this matters: the graph does not grow with frames. It grows with *identity and interaction*, and trajectories are bounded (Section 4).

3 Perception: spend compute only where it matters

1. **Motion gate.** Mask dynamic boxes and hands, estimate a partial affine transform between frames, and classify the camera as stationary or panning. Below 2.5 px the scene is still; above 8 px it is a pan. This separates head rotation from real object motion.
2. **Detection.** YOLO-World is open-vocabulary but *prompt-driven*: it only labels objects it is asked about. The starter list covers 12 objects; an opt-in broad preset covers 98 common indoor/egocentric objects.
3. **Hands.** MediaPipe returns 21 landmarks. Contact is a geometric test, not a guess.
4. **Tracking & Re-ID.** ByteTrack supplies kinematic IDs; inactive tracks are re-bound by CLIP appearance similarity (≥ 0.82). Graph-level consolidation merges legacy duplicates without merging co-visible objects.
5. **Interaction test.** Intersection over Smaller Area:

$$\mathrm{IoS}(A, B) = \frac{\mathrm{Area}(A \cap B)}{\min(\mathrm{Area}(A), \mathrm{Area}(B))}$$

   Interaction fires when a fingertip enters the object hull, or IoS(hand, object) > 0.60 for ≥ 0.3 s.
6. **Temporal strip.** Keep a 30-frame RAM ring buffer. On an interaction, stitch three moments (pre / contact / post) into one 1×3 image and ask Florence-2 for an action caption. Strict JSON parsing and per-action thresholds reject hedged or implausible captions.

Constant	Value used by the implementation
Still / pan threshold	< 2.5 px / > 8 px affine magnitude
Re-ID cosine	0.82
Place-recognition cosine	0.85
Interaction / containment IoS	0.60 / 0.85
Interaction dwell	0.30 s
Temporal buffer	30 frames; strip panels 336 px
Lost / trajectory retention	600 s / 600 s
Verification queue	depth 8; dedup window 1024 clips
Language model	`qwen2.5:7b-instruct`; 4096 ctx; 30 min keep-alive

3.1 Object appearance attributes

Every Nth detection crop gets a cheap, deterministic appearance pass; no model download and a few milliseconds per box on CPU. It answers the question a label alone cannot: *which* cup?

- **Dominant colour.** Trim 10% from every crop edge, resize the inner region to ≤ 64 px, and run 3-means in BGR. Cluster weights are a Gaussian centre prior multiplied by a background-suppression term $\mathrm{clip}(\|c-b\|/80,\ 0.15,\ 1)$, where b is the median border colour. The central cluster wins when it covers $\geq 12\%$ of the crop. The primary BGR centroid becomes a stable colour word plus a `#rrggbb` swatch.
- **Colour word.** Achromatic pixels are classified first ($V < 42$: black; $S < 35$: white/grey), then hue bands map to red, orange, yellow, green, cyan, blue, purple or pink.

- **Shape.** Bounding-box aspect ratio: ≤ 0.82 tall, ≥ 1.22 wide, otherwise square.
- **Size.** Relative crop area: $\leq 1.2\%$ small, $\leq 7.5\%$ medium, otherwise large.
- **Brightness / saturation.** $V < 70$ dark, $V < 145$ dim, else bright; $S < 55$ muted, $S < 150$ colourful, else vivid.
- **Texture / position.** Laplacian variance ≥ 45 is detailed, else plain; the box centre gives left/centre/right and top/middle/bottom.
- **Stability.** A new colour replaces the track colour only when its confidence is at least 0.12 higher; geometry and position update immediately.

The flattened attributes are persisted on the foreground entity and exposed in snapshot/CSV exports, graph tooltips and GraphRAG. Queries such as *"what colour is the mug?"*, *"how big is it?"* and *"describe the backpack"* cite the entity's last observation with relation `APPEARANCE`; a missing attribute is answered as ungrounded rather than guessed. The offline evaluation set includes colour, size, describe and missing-detail cases and currently passes 13/13 with 100% citation precision.

4 Storage: why the graph is small

SQLite is a durable projection of the in-memory NetworkX graph, not a second copy. WAL mode, `synchronous=NORMAL`, foreign keys, and indexes keep reads non-blocking: `(src, relation)`, `(dst, relation)`, `(t_start)`, `(entity_id, t_event)`.

- **Embeddings are per identity, not per frame.** A 512-d CLIP vector is stored once per entity/zone as float32 (2 KB), not for every observation.
- **Observations are trajectories only.** Bounding boxes and state, no pixels. Rows older than 10 minutes are purged. Anchors are never deleted.
- **Zones, furniture, transitions, edges, anchors and observations** are separate indexed tables; the schema is versioned in `meta`.
- **Export is portable.** `snapshot.json` contains graph, entities, actions, transitions, furniture, verification telemetry and backends; CSVs use stable headers.

Measured on this repo (45 s Aria kitchen clip)	Value	Unit
Source video	2.99	MB
Graph database, including WAL	0.436	MB
Entities / edges / transitions / anchors	11 / 53 / 214 / 28	count
Bounding-box observations	536	rows
Entity embeddings	22	KB
Naive per-observation 512-d embeddings	1.05	MB
Deterministic query latency	0.14–1.05	ms
Exported snapshot	190	KB

Scale rule. Storage is $O(\text{identities} + \text{interactions})$, not $O(\text{frames})$. Trajectories are bounded by the 10-minute window; state transitions and terminal anchors are retained. On very long videos, transitions still grow, so retention is a policy decision, not a free lunch.

5 Query: graph first, language second

1. **Resolve** the entity (aliases, plurals, ambiguity-aware scoring) and the active zone. The result carries `match_score`, `confidence`, `match_reason` and ranked `candidates`.
2. **Serialize** the 2-hop subgraph: latest state, spatial anchor, last interaction, parent zone, and the relevant action edges.
3. **Synthesize** either a deterministic template (always available) or prose from local Qwen through Ollama. The model sees only the table, never the video.
4. **Validate** every citation. A timestamp that does not map back to the subgraph is a hallucination and is dropped; if no valid citation remains, the answer falls back to the template.

Example. *"Where are my keys?"* → `The keys are resting on the dining table in the kitchen [t=16.4s].` The citation is clickable in the dashboard and seeks the video to 16.4 s. The newest grounded citation can also attach a supporting still frame (interaction strip, crop, zone keyframe, or an extracted video frame).

Why a small model is enough. The hard work is retrieval and grounding. The LLM's job is phrasing. This is also why the system survives with no model at all: the deterministic answerer uses the same subgraph.

6 Interfaces and operations

Surface	What it is for
`chronicle sources`	List bundled, real and uploaded clips.
`chronicle run`	Ingest a video through perception, memory and semantics.
`chronicle ask`	Query a persisted SQLite memory from the terminal.
`chronicle serve`	Operator dashboard: live graph, timeline, tables, log, GraphRAG chat.
`run_dashboard.py`	Mission-control dashboard: time travel, overlays, inspector, query console.
`chronicle.sh`	Start/stop/restart/status/logs/tests; LLM flags.

```
chronicle sources
chronicle run --source synthetic
chronicle ask "where are my keys?" --db data/demo/chronicle.db
chronicle serve --source synthetic --auto-run
./chronicle.sh start --llm-model qwen2.5:7b-instruct
./chronicle.sh start --no-llm
python run_dashboard.py --source demo
python -m pytest -q
```

Uploads. Drop a clip into the operator dashboard. The targets field controls the YOLO-World prompts (comma separated); *Broad scene* enables the 98-object preset. The chosen vocabulary is persisted in the upload's `source.json` manifest, and every upload gets its own SQLite memory.

6.1 Configuration cheat sheet

Variable	Default / meaning
`CHRONICLE_LLM_BACKEND`	`auto`, `ollama`, `transformers`, `template`, `none`
`CHRONICLE_LLM_MODEL`	`qwen2.5:7b-instruct`
`CHRONICLE_OLLAMA_URL`	`http://127.0.0.1:11434`
`CHRONICLE_QUERY_URL`	Optional external answer endpoint
`CHRONICLE_DETECTOR`	`auto`, `color`, `yolo-world`
`CHRONICLE_HANDS`	`auto`, `color`, `mediapipe`
`CHRONICLE_EMBEDDER`	`auto`, `color`, `deterministic`, `openclip`
`CHRONICLE_SOURCE`	`demo`, `live`, `jsonl`, `sqlite`
`CHRONICLE_DATA_DIR`	Asset root (default `./data`)

7 Quality: what is actually verified

- **Bounded verification queue.** Size 8, same-clip dedup, drop-oldest or reject policy, depth/backpressure/latency telemetry. VLM latency cannot stall ingestion.
- **Calibration.** Labelled captions produce per-action thresholds instead of one global confidence knob.
- **Action-timeline evaluation.** Greedy nearest-time matching against labelled actions; precision, recall, timestamp error, entity/container accuracy.
- **GraphRAG evaluation.** Location, time, action, removal, lost, zone-only and unknown cases; citation precision is checked per answer.
- **Tests.** 250+ model-free tests cover graph storage, lifecycle, query frames, snapshots, uploads, timeline evaluation and the dashboard smoke path.

8 Research grounding (and what this implementation adds)

Work	Idea borrowed	Engineering delta here
R4DSG (ACM MM 2026)	Anchor-relative 4D memory; persistent objects; no global 3D model	Detection + tracking instead of segmentation/3D lifting; SQLite projection; 0.4 MB per clip.
VL-KnG (2026)	Persistent spatio-temporal KG; graph retrieval decouples latency from video	Millisecond deterministic queries; optional small LLM; bounded queue; no LLM-based association.
ESOM (WACV 2026)	Streaming object memory and online identity	30-frame ring buffer, inactive Re-ID memory, graph consolidation; no visual query localization.
Hand-object contact (WACV 2024)	Contact-based interaction triggering	Faithful IoS/hull + 0.3 s dwell implementation with explicit JSON action validation.
LightMem-Ego	Working / episodic / archival tiers	Ring buffer → foreground graph/SQLite → terminal anchors and zones.

Honest positioning. The research novelty is in the cited papers. Chronicle's contribution is a compact, offline, auditable *system*: small graph, fast retrieval, explicit citations, and a UI that shows the evidence instead of asking for trust.

9 Ten-minute demo script

1. Start on the synthetic fixture. Show the graph, timeline and live pipeline log.
2. Ask *"Where are my keys?"* Click the citation: the video seeks to the moment.
3. Scrub back before the placement and ask again. The answer changes. That is episodic memory, not a transcript.
4. Switch to the HoloLens or Aria clip. Ask a multi-hop question: *"Where were the keys before the drawer?"*
5. Show the verification queue, calibration numbers and citation precision.
6. Stop Ollama or run `-no-llm`; repeat a query. The deterministic answer still cites the graph. Robustness is part of the demo.
7. Export `snapshot.json` and a CSV to show the memory is portable.

Pitfalls to rehearse. (i) YOLO-World only sees prompted labels — use *Broad scene* or name the objects. (ii) Preload Qwen before going on stage; cold 7B load is dead air. (iii) Checkpoint WAL before quoting database size: `PRAGMA wal_checkpoint(TRUNCATE)`. (iv) Verify evidence images exist for real clips; otherwise the frame falls back to video extraction. (v) Do not claim Florence-2/OpenCLIP unless those extras are installed on the demo machine.

A File map

Path	Role
`chronicle/perception/`	Motion gate, detector, hands, tracker, interaction geometry.
`chronicle/graph/`	Foreground/background graphs, state machine, query engine, consolidation.
`chronicle/storage/`	SQLite schema and durable projections.
`chronicle/semantic/`	VPR, temporal strips, Florence-2, actions, queue, GraphRAG, LLM.
`chronicle/integration.py`	Runtime facade used by CLI, dashboards and demos.
`dashboard/`	Mission-control dashboard, query console, API, evidence frames.
`chronicle/ui/`	Operator dashboard (NiceGUI), uploads, snapshot import/export.
`docs/`	Contracts, role handoff notes, evaluation examples.

B Glossary

IoS
Intersection over Smaller Area; containment-friendly alternative to IoU.

VPR
Visual place recognition; CLIP keyframe matching for zone relocalization.

Re-ID
Re-identification; appearance matching that re-binds a lost track to an entity.

Terminal anchor
The last verified spatial relation of an object; never deleted.

GraphRAG
Retrieve a 2-hop subgraph, then synthesize a grounded answer.

Evidence frame
The still image that supports an answer: interaction strip, crop, zone keyframe, or extracted video frame.